MMN surface preparation of Titanium and how it affects osseointegration.

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Titanium implants make use of additional surface preparation in order to promote osseointegration, which can significantly decrease the time required for a patient's recovery. The premise of what features a surface preparation may impart on an implant and how those features are desirable for improving osseointegration are explored, primarily focusing on increasing surface free energy to promote protein adsorption and cellular adherence. Several methodologies for surface preparation of titanium implants are considered, weighing their benefits and downfalls to comment on their efficacy. Further methods for increasing bioactivity of the implant and manufacturing that may promote improved surfaces are also discussed.

Introduction

Titanium is among the most commonly used materials for medical implants due to its strong mechanical properties, excellent biocompatibility, and exceptional osseointegration, bonding with the bone structure that grows upon it. The conditions under which osseointegration occurs best have been observed and experimented with across a number of alloys, surface finishes, and surface finishing techniques, intending to alter the initial conditions upon which mesenchymal stem cells begin developing into organized osteoblast matrices on the surface of the implant. Alloys are used to improve the base material properties generally to extend the fatigue life of the implant or improve biocompatibility even further. Surface finishes contribute significantly to the ability of the osteoblasts' ability to adhere to the implant, and even the technique with which a surface is applied can impart additional properties onto it that affect the osseointegration process. The methods of interest to this paper are those making use of plasma, acid etching, electro-deposition, traditional grinding and grit blasting, and laser processing. Additionally, there exist some techniques that expand beyond using just the base titanium alloy, making use of ceramic top coatings to increase the implant's ability to bond to the bone.

Macro Micro Nano Surfaces Background

There are a number of reasons as to why altering the surface finish improves osseointegration. The most evident is that increasing the roughness of the surface increases the surface area and therefore the area upon which osteoblasts are able to adhere to. Another contributing factor altered by the surface finish is the surface

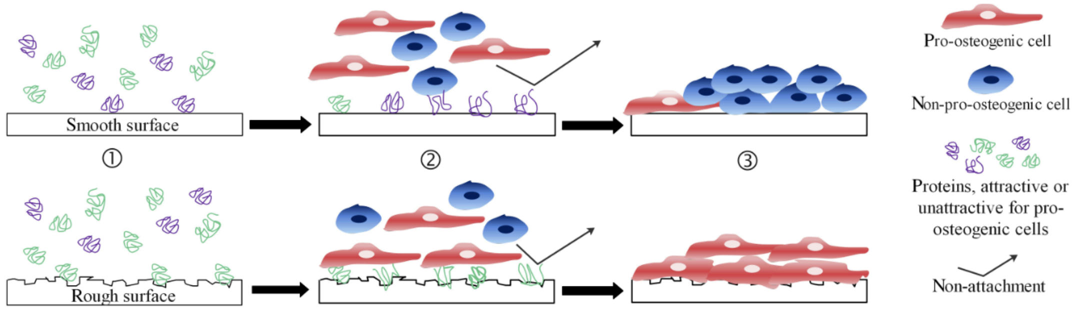

Fig 1. Depiction of cell-material interaction for smooth and rough surfaces. [from "Implant-bone-interface: Reviewing the impact of titanium…" by Stich, Alagboso, Krenek]

free energy, directly affecting the ability for a matrix to nucleate on the implant's surface.

Surface free energy (SFE) is a measure of the unsatisfied, open bond energy on a material's surface. Having higher SFE on a specimen results in a pull across the surface to fill those open bonds, increasing its wettability when interacting with a fluid. The contact angle formed by a drop of fluid with known properties on a surface is both directly affected by the SFE, and used as a common method for measuring a surface's wettability. Lower contact angles align with a high wettability and SFE. For implants, this pulls water and blood across the surface more quickly and with stronger adhesion, improving the initial steps of osseointegration.

Another benefit of high SFE is in its effects on protein adsorption. Protein adsorption is desirable, as it forms an initial layer on the surface upon which osteoblasts adhere on top of. Although proteins do adsorb under hydrophobic conditions, proteins orient themselves in a more effective way when adhering to a hydrophilic surface. This leads to a stronger interlayer between the osteoblasts and surface, increasing the surface area in which ossification can occur and improves the adherence of the organic cells to the implant.

Additional reasonings were discussed across the resources used as to why different roughnesses may improve osseointegration. One idea discussed and supported is that by emulating the surface of natural bone in the implant through biomimetic surface architecture, the cells will recognize this similarity and treat it as if it were natural bone. This would result in a faster adoption of the implant in the body, and an improved biocompatibility.

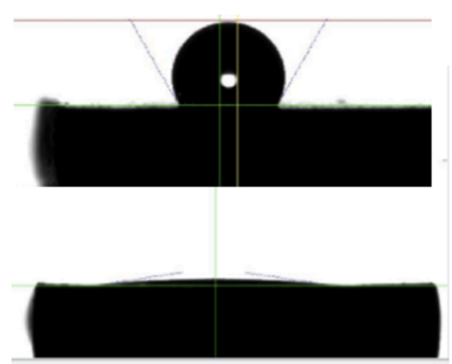

Fig 2. Example of contact angles between a treated and untreated surface. [from "Different Methods to Modify the Hydrophilicity of Titanium…" by Jacobs, Dillon, Cohen, Boyan, and Schwartz.]

Titanium & its Alloys

The primary intent of altering the alloy used in an implant is not to improve in osseointegration or biocompatibility, but rather to improve its mechanical properties. Generally, the top layer of any titanium alloy forms a TiO_2 layer from surface passivation with the air. This uniformity in the top surface's composition across titanium and its alloys results in near-ubiquitous osseointegration and biocompatibility. Additionally, the oxide layer allows for comparison of surface preparation methods across different alloys of titanium, as they share the same surface properties regardless of their alloying elements.

Surface Preparation Methods

A number of surface preparations can be applied to increase SFE, surface area, and improve osseointegration. These alter the surface using mechanical, electrochemical, and thermal means.

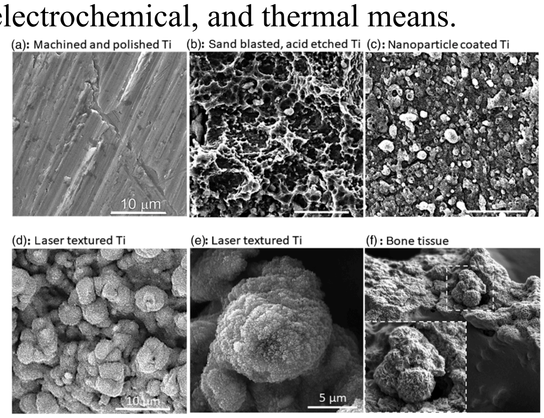

Fig 3. Comparison of different surface textures finishes at 10 µm. (a) Machined and Polished, (b) Sand-blasted & acid etched, (c) Nanoparticle coated, Laser textured at (d) 10 µm and (e) 5 µm, (f) Bone tissue. [from "Implant-bone-interface: Reviewing the impact of titanium…" by Stich, Alagboso, Krenek]

Mechanical Preparation

Gritblasting

Gritblasting operates by impacting the surface with particles propelled with pressurized air. These particles scratch the surface, and as they cut through may become stuck. Because of this, it is of great importance that the particles used are biocompatible and do not inhibit proteins or cell formations. By cutting into the surface, its surface area and free surface energy is increased, improving the conditions for protein and cell formation.

As a result of particles becoming stuck in the surface, gritblasting is one of few methods that can alter the composition of the surface layer. TiO_2 particulates can be used, in which case the composition remains unchanged, but there also exists both particulates that can inhibit or encourage bioactivity at the surface. By using bioactive materials like bioceramics that simulate real bone, improved nucleation sites can be imparted to the surface to improve the beginnings of osseointegration. Gritblasting can also be used in conjunction with other methodologies with little to no interference, compounding the benefits of its increased free surface energy and nucleation sites with the results of other methods.

Chemical & Electrochemical Preparation

Acid Etching

Acid etching can be used to create micropits of sub-micron to several micron depth, improving the surface layer, however on its own it's a somewhat flawed methodology. Acids used to attack the oxide layer are able to dissolve it, however in the process they may impart hydrogen into the structure of the titanium. This bonds with the titanium to create titanium hydride, which is far more brittle than the other compounds encountered, weakening the implant. By finely controlling the variables at play, primarily acid concentration, exposure time, and solution temperature, one may be able to tune a consistently beneficial method that avoids embrittlement.

When used in conjunction with gritblasting, acid etching may be used to dissolve the grit contaminants that remain lodged in the surface, further refining the surface. This usage as a more post-process method is reliable and more widely used.

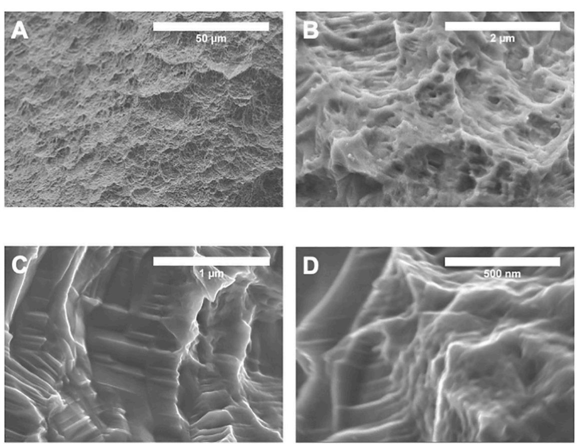

Fig 4. Gritblasted & acid etched surface finish from 50 µm to 500 nm. [from “Growth Factors Produced by Bone Marrow Stromal Cells…” by Berger, Bosh, Jacobs]

Electrochemical Deposition

Electrochemical deposition (ECD) uses the titanium implant as a cathode submerged with an anode in an electrolyte solution, containing whatever particle is desired to be deposited suspended in it. When current is applied, the particles are attracted through the solution to the titanium implant, forming an even coating across the surface of the implant. Varying the current can produce the rougher deposition surfaces desired for osseointegration. ECD’s uniform results and consistency make it an extremely valuable methodology for increasing surface roughness, and it can be used similarly to gritblasting with more bioactive compounds to promote nucleation.

Anodization

Anodization is the inverted process of electrochemical deposition. The implant is submerged in an electrolyte solution and instead acts as the anode, with an inert metal acting as the submerged electrode. As the electricity flows through the system, the water molecules of the solution split and release oxygen, which then bonds with the surface of the titanium implant. This forms a layer of titanium oxide, and can be controlled similarly to electrochemical deposition by altering the applied voltage.

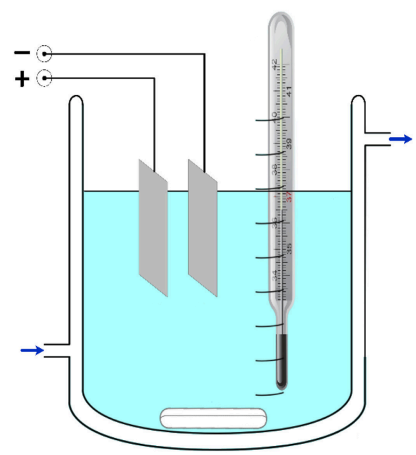

Fig 5. Depiction of anode, cathode, and electrolyte system for use in either electrochemical deposition or anodization. [Adapted from “Evolution of anodised titanium…” by Alipal, Lee, Koshy.]

Of particular note is that the electrolyte can be altered, and by using fluoride or chloride based electrolytes titanium nanotubes can be developed (TNT). TNTs form as the electrolyte simultaneously creates and dissolves the titanium oxide layer, producing self-ordered nanotubes. These nanotubes can be developed to mimic the porosity and structure of bone cells, forming biomimetic surface architectures that are beneficial to osseointegration.

Thermal Preparation

Direct Laser Interference Patterning

Direct Laser Interference Patterning (DLIP) makes use of multiple lasers aimed at the titanium surface. These lasers are forced to interfere with each other periodically, creating patterned structures across the surface featuring high and low laser intensities that ablate and melt the surface. This induces varying heights and crevices across the surface of the implant, leading to the desired high surface energy and increased surface area for nucleation.

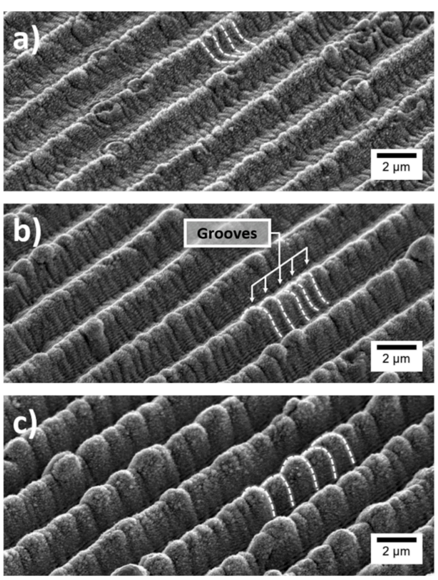

Fig 6. Tilted SEM imagery of DLIP microstructured grooves. [from "Fabrication of four-level hierarchical topographies…" by Schell, Alamri, Hariharan]

One benefit of DLIP is that it is exceptionally fast, performing each operation in typically either nano or pico second pulses. This means DLIP can apply its surface finish both uniformly and reliably due to its patterning, and quickly, making it an ideal process for bulk processing.

Plasma Treatment

Plasma treatments operate by exposing the surface of the titanium to ionizing gases, pulling away hydrocarbons from the surface. These hydrocarbons are non-polar, and contribute greatly to the natural hydrophobicity of the surface. Without them, hydrophilicity and free surface energy is increased, doing so without removing any of the base oxide layer or imparting deformation to the surface.

Plasma treatments are unfortunately temporary, as air exposure results in atmospheric hydrocarbons rapidly adsorbing back into the specimen and reverting it back to its original hydrophobic state. Therefore to be effective, plasma treatments must be applied to a specimen and then implanted as quickly as possible to minimize its exposure to hydrocarbons.

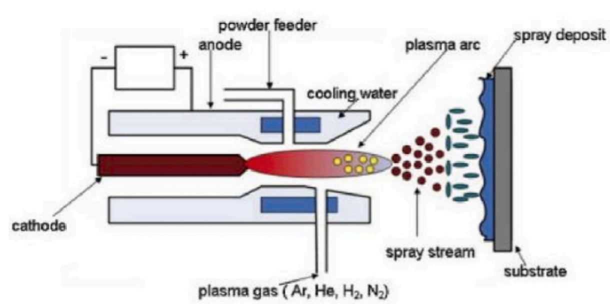

Fig 7. Example of a plasma spraying schematic. [from "Advanced surface engineering of titanium…" by Jiang, Zhang, Hu]

Beyond Titanium

An additional way to improve an implant's osseointegration is to forgo a titanium surface entirely. By making use of electrodeposition or other methods mentioned above, a different material coating, such as a bioceramic, may be applied across the entire surface of the implant. This greatly increases the ability for cells to form matrices that interface with the implant. Depending on the material, this method could also be used to better produce biomimetic surface architecture and emulate the body attempting to repair real bone.

Concluding Statements

The methodologies used to improve titanium implant osseointegration through surface preparation each present their own distinct advantages and tradeoffs to achieve the same objective of increasing the surface energy, the surface area, or promoting protein adsorption. Mechanical approaches like gritblasting offer versatility when combined with other methods, and the ability to impart bioactive particles to the finished surface. Electrochemical deposition provides superior uniformity and reliability, producing consistent coating thickness and roughness. Thermal methods such as DLIP enable precise, rapid processing, whereas plasma treatments are able to achieve the desired hydrophilicity and free surface energy without any surface damage. The selected surface preparation method should thus be informed by the implant design, manufacturing scale, and context of the situation. Additionally, synergistic approaches that make use of several methodologies may bridge the downfalls the individual methods face, enabling methods to improve in efficacy when combined with others. As research continues refining and exploring surface preparation techniques, the ability of the surface to promote osseointegration and improve both recovery time and implant stability will ultimately continue to enhance patient outcomes.

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